

Rank Measures and Exponential Lower Bounds for Multilinear Secret Sharing

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Abstract

A multilinear secret-sharing scheme shares a vector secret and can therefore amortize share size over the secret dimension. This amortization can invalidate lower bounds proved for one-dimensional linear schemes, and the best previous explicit lower bound for multilinear schemes was quasipolynomial, $n^{\Omega(\log n)}$. We prove that the Razborov–Gál rank measure survives amortization: the normalized size of a multi-target monotone span program is at least the rank measure of the function it computes. Combined with the rank witnesses of Pitassi and Robere, this gives an explicit family of access structures for which every perfect multilinear scheme over every finite field has average and maximum information ratio $2^{\Omega(n)}$. The worst-case multilinear information ratio is therefore $2^{\Theta(n)}$, answering a question of Beimel. We further extend the bound to schemes whose sharing algorithm is arbitrary and whose reconstruction is affine-linear, under pairwise statistical privacy below one; combined with the degree-reduction theorem of Beimel, Othman, and Peter, this yields exponential normalized lower bounds for every fixed reconstruction degree whenever the secret dimension is $2^{o(n)}$.

1 Introduction

A secret-sharing scheme distributes a secret among a set of n participants. The authorized sets can recover the secret, whereas the unauthorized sets should obtain no information about it. The family of authorized sets is called an access structure. If the secret consists of d_0 field elements and participant i receives d_i field elements, then the *information ratio* of participant i is $\rho_i = d_i/d_0$. We write

$$\rho_{\max} = \max_i \rho_i, \quad \rho_{\text{av}} = \frac{1}{n} \sum_i \rho_i, \quad \rho_{\text{tot}} = \sum_i \rho_i = \frac{d}{d_0},$$

where $d = \sum_i d_i$, for the maximum, the average, and the total information ratio of the scheme. The first two are the standard measures of efficiency.

In a scalar linear scheme the secret is one field element and every share is a linear function of the secret and of fresh randomness. A multilinear scheme shares a vector of d_0 field elements the same way. Allowing d_0 to grow is a form of amortization: the same access structure can sometimes be realized with much smaller share dimension per secret dimension, and there are access structures where exactly this happens [5, 9]. A lower bound proved for $d_0 = 1$ therefore need not survive, because the quantity being bounded changes with the secret dimension.

For $d_0 = 1$ the model is that of monotone span programs, and exponential lower bounds are known: Robere, Pitassi, Rossman, and Cook proved them over the reals [26], and Pitassi and

Robere over every field [24]. Both go through the Razborov–Gál rank measure. For multilinear schemes the best explicit bound was quasipolynomial, $n^{\Omega(\log n)}$, obtained by Beimel, Ben-Efraim, Padró, and Tyomkin (BBPT) by extending a unique-intersection rank argument to multi-target monotone span programs [7]. Beimel’s recent monograph records this state of affairs and asks whether the worst-case multilinear information ratio is exponential [9, Question 10.4]. We answer that question.

The obstruction we transport is the Razborov–Gál rank measure. Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$ be monotone, let $U = f^{-1}(1)$ and $V = f^{-1}(0)$, and let A be a matrix over a field $\mathbb{F}$ with rows indexed by U and columns by V . Participant i *separates* a pair $(u, v) \in U \times V$ when $u_i = 1$ and $v_i = 0$; write R_i for the set of such pairs and $A|_{R_i}$ for the matrix obtained by setting all entries outside R_i to zero. The measure is

$$\mu_{\mathbb{F}}(f) = \max_{A \neq 0} \frac{\text{rank}_{\mathbb{F}} A}{\max_i \text{rank}_{\mathbb{F}}(A|_{R_i})}.$$

The numerator measures a global obstruction; the denominator measures the largest part of that obstruction visible to a single participant.

A multi-target monotone span program (MTMSP) is the linear-algebraic representation of a multilinear scheme. Its rows are labeled by participants; an authorized set must span a d_0 -dimensional target space, and an unauthorized set must meet that space trivially. Participant i labels exactly d_i rows when its share has dimension d_i . Our main result is proved in section 3.

Theorem 1.1 (Amortized rank-measure theorem). *Let a d_0 -target MTMSP over $\mathbb{F}$ compute f , and suppose that participant i labels d_i rows. Then every nonzero matrix $A \in \mathbb{F}^{U \times V}$ satisfies*

$$d_0 \text{rank } A \leq \sum_{i=1}^n d_i \text{rank}(A|_{R_i}).$$

Maximizing over A gives

$$\frac{\sum_i d_i}{d_0} \geq \mu_{\mathbb{F}}(f).$$

Combining the theorem with the characteristic-separated rank witnesses of Pitassi and Robere gives the following field-uniform consequence; see theorems 4.6 and 4.7.

Theorem 1.2 (Exponential multilinear lower bound). *There are an explicit family (Γ_n) and constants $\alpha > 0$ and n_0 such that, for every finite field $\mathbb{F}$ and every $n \geq n_0$, every perfect multilinear scheme for Γ_n satisfies*

$$\rho_{\text{av}} \geq \frac{2^{\alpha n}}{n} \quad \text{and} \quad \rho_{\text{max}} \geq \frac{2^{\alpha n}}{n}.$$

Consequently, the worst-case multilinear information ratio is $2^{\Theta(n)}$.

The same method also applies when the dealer’s sharing algorithm is arbitrary. Beimel, Othman, and Peter observed that affine-linear reconstruction vectors can be arranged as an MTMSP for the dual access structure [13]. We give the precise affine version of this construction and combine it with the main theorem.

Theorem 1.3 (Affine-linear reconstruction). *Let a perfectly correct scheme over a finite field $\mathbb{F}$ have arbitrary sharing and affine-linear reconstruction. Suppose that the differences of the allowed secrets span $\mathbb{F}^{d_0}$ and that the pairwise statistical privacy parameter is smaller than one. If all shares together have dimension d , then*

$$\rho_{\text{tot}} = \frac{d}{d_0} \geq \mu_{\mathbb{F}}(\Gamma).$$

Finally, the degree-reduction theorem of Beimel, Othman, and Peter converts degree- r reconstruction into affine-linear reconstruction with a controlled polynomial increase in share dimension. For every fixed r , the resulting bound is exponential whenever the secret dimension is $2^{o(n)}$. This gives a partial answer to their question on normalized share size for long secrets and bounded-degree reconstruction [13, Question 1.1]; arbitrary secret dimensions remain open when $r > 1$.

In summary, the paper contains four results.

- (1) Theorem 3.1 (section 3): the Razborov–Gál rank measure lower-bounds the *normalized* size of a multi-target monotone span program, $\text{amSP}_{\mathbb{F}}(f) \geq \mu_{\mathbb{F}}(f)$.
- (2) Theorem 4.7 and corollary 4.9 (section 4): an explicit family (Γ_n) whose multilinear information ratio is $2^{\Omega(n)}$ over every finite field, whence the worst-case multilinear information ratio is $2^{\Theta(n)}$. This answers [9, Question 10.4].
- (3) Theorem 5.6 (section 5): the same lower bound for schemes with arbitrary sharing, affine-linear reconstruction, and pairwise statistical privacy parameter below one.
- (4) Theorem 6.2 (section 6): for every fixed reconstruction degree r and every secret dimension $2^{o(n)}$, an exponential normalized lower bound; a partial answer to [13, Question 1.1].

1.1 What is new here, and what is not

The hard access structures are not new. They are the characteristic-separated families of Pitassi and Robere [24], the same objects that already gave $2^{\Omega(n)}$ for scalar linear schemes. There is no new hard function here, no new proof-complexity input, and no new construction. The only modification we make is to take the disjunction of two of their families on disjoint variable sets, so as to obtain a single family that is hard over every finite field; the corresponding statement for $d_0 = 1$ is already recorded in the literature [13, extended version, Theorem 4.10]. What changes is one step in the argument, and that step is what lets these witnesses be fed into the multi-target model at all.

Where the barrier actually was. The barrier was never that amortization is powerful and multilinear bounds must therefore be weaker. The rank measure is a ratio, and BBPT’s multilinear bound runs through the Gál–Pudlák unique-intersection criterion, which is the special case in which every participant restriction $A|_{R_i}$ has rank one: the denominator is trivially 1 and all the work is done by the numerator. Even for one-target programs, that criterion has never yielded more than $n^{\Omega(\log n)}$; the quasipolynomial ceiling was already there in the scalar case [17, 18, 26]. BBPT were therefore not losing to amortization. They were losing to the criterion. Nor would handing them the Pitassi–Robere functions have helped: those witnesses are exponential precisely because their participant restrictions have *large* rank, which is exactly the information that the unique-intersection framing discards.

Why a general witness was hard to use with many targets. With a single target, the certificates attached to a separated pair (u, v) are scalars $b_i[u, v]$ summing to 1; one multiplies through by the witness entry $A[u, v]$ and applies the Hadamard-product rank inequality directly. With d_0 targets, simultaneous reconstruction and simultaneous rejection produce $d_0 \times d_0$ matrices $B_i[u, v]$ summing to the identity. A scalar witness and a matrix-valued tiling do not obviously combine, and it is not a priori clear that the denominator of $\mu_{\mathbb{F}}$ survives the passage.

The step that makes them combine. Flatten the tiles, so that B_i becomes a matrix with rows indexed by $U \times [d_0]$ and columns by $V \times [d_0]$. Weighting the block (u, v) by the scalar $A[u, v]$ is then exactly the Hadamard product with $A|_{R_i} \otimes J_{d_0}$, where J_{d_0} is the all-ones matrix. Over every field $\text{rank } J_{d_0} = 1$, so

$$\text{rank}(A|_{R_i} \otimes J_{d_0}) = \text{rank}(A|_{R_i}),$$

and the weighting costs nothing. But the very same weighted tiles sum to $A \otimes I_{d_0}$, and $\text{rank}(A \otimes I_{d_0}) = d_0 \text{rank } A$. That asymmetry — the all-ones matrix on the weighting side, the identity on the summation side — is the whole mechanism. It produces

$$d_0 \text{rank } A \leq \sum_{i=1}^n d_i \text{rank}(A|_{R_i}),$$

in which the factor d_0 on the left is exactly what cancels the amortization. The other half of the proof is that $\text{rank } B_i \leq d_i$ comes from a single *global* factorization through participant i 's d_i rows, valid simultaneously for all pairs (u, v) , and not from a blockwise count. Remark 3.5 states the comparison with BBPT precisely.

In one sentence. The obstruction that was believed to be lost under amortization is not lost: a single factorization argument carries the full Razborov–Gál measure to arbitrary target dimension, and the exponential witnesses already in the literature then apply to the multilinear model verbatim.

Two things the theorem does not say. First, it does not say that amortization is useless. Applebaum and Arkis exhibit access structures for which long secrets genuinely reduce normalized share complexity [5], and those separations are against *counting* bounds [9]. Our claim is narrower and correspondingly sharper: hardness that is witnessed by a rank measure cannot be amortized away. Second, the argument consumes perfect privacy in the form of strong rejection, $\text{rowspan}(M_B) \cap T = \{0\}$. It says nothing about ramp or leaky forbidden sets; see remark 3.8 and section 7.

1.2 Related work

General access structures. Secret sharing was introduced independently by Shamir [28] and Blakley [14]. Classical monotone-formula constructions give exponential upper bounds for arbitrary access structures. On the lower-bound side, Csirmaz proved an $\Omega(n/\log n)$ maximum information-ratio bound [16], and Williamson recently obtained linear maximum-share lower bounds from Shannon inequalities [29]; both concern *unrestricted* schemes and are therefore complementary to the present work rather than competing with it, since no linearity is assumed there and the bounds are polynomial rather than exponential. On the upper-bound side, Liu and Vaikuntanathan broke the circuit-size barrier [21], and the general exponent was subsequently brought down to $1.5^n = 2^{0.585n+o(n)}$ by Applebaum and Nir [6], with [1, 3, 12, 2, 4] contributing further improvements for general, uniform, slice, and linear schemes along the way; Beimel and Farràs determined the share size for almost all access structures [10]. Most recently Nir gave a linear scheme realizing every n -party access structure with a one-bit secret and maximum share size $2^{\lceil n/2 \rceil - 1} + 1$, which matches the counting lower bound of $2^{n/2 - o(n)}$ for monotone span programs and thereby fixes the worst-case *linear* exponent at $1/2$; combining a variant of that scheme with a nonlinear scheme of Applebaum, Beimel, Nir, and Peter he also lowered the general exponent to $2^{0.496n+o(n)}$, separating the worst-case linear and nonlinear exponents [22]. The multilinear exponent studied here is at most the linear one and at least the general one; corollary 4.9 places it in $[\alpha, \frac{1}{2}]$.

Linear schemes and rank methods. The correspondence between linear secret sharing and monotone span programs goes back to Karchmer and Wigderson [20]; see also the surveys [8, 9]. Rank methods for span programs were developed by Razborov [25], Gál [17], and Gál and Pudlák [18]. Robere, Pitassi, Rossman, and Cook obtained the first exponential span-program lower bounds through the rank measure, over the reals [26], and Pitassi and Robere’s lifting theorem extended this to every field [24]; the latter supplies the witnesses used here. BBPT introduced the multi-target model and proved the first explicit superpolynomial lower bound for multilinear secret sharing [7].

Amortization and polynomial reconstruction. Applebaum and Arkis showed that long secrets can substantially reduce normalized share complexity [5]. Polynomial secret-sharing schemes were studied in [23, 13, 11]. Beimel, Othman, and Peter proved the degree-reduction theorem used here and asked for normalized lower bounds for long secrets with fixed reconstruction degree [13].

1.3 Organization

Section 2 introduces secret sharing, MTMSPs, and the rank measure. Section 3 proves the amortized rank theorem. Section 4 derives the field-uniform exponential lower bound. Section 5 treats arbitrary sharing with affine-linear reconstruction. Section 6 gives the bounded-degree consequence and the conditioning lemma, and section 7 concludes with open problems.

2 Preliminaries

Convention 2.1 (Constants and uniformity). All asymptotic statements are with respect to the number n of participants or Boolean inputs. Constants implicit in O , Ω , and Θ may depend on a fixed prime or a fixed reconstruction degree when this is indicated, but they do not depend on n or on the target dimension d_0 . When a statement ranges over all finite fields, both the hidden constant and the threshold are independent of the field. The constant K_r used for degree- r reconstruction depends only on r .

2.1 Access structures and secret-sharing schemes

Let $P = [n] = \{1, \dots, n\}$ be the participant set. A family $\Gamma \subseteq 2^P$ is *monotone* if $A \in \Gamma$ and $A \subseteq B$ imply $B \in \Gamma$. An *access structure* is a nonempty monotone family with $\emptyset \notin \Gamma$. Sets in Γ are *authorized*; the other sets are *unauthorized*. The monotone Boolean function associated with Γ is

$$f_\Gamma(x) = 1 \iff \{i : x_i = 1\} \in \Gamma.$$

The dual access structure is

$$\Gamma^* = \{A \subseteq P : P \setminus A \notin \Gamma\}.$$

A secret-sharing scheme consists of a finite secret domain $\mathcal{S}_0$, finite share domains $\mathcal{S}_1, \dots, \mathcal{S}_n$, and a randomized sharing map

$$\text{Share} : \mathcal{S}_0 \longrightarrow \mathcal{S}_1 \times \dots \times \mathcal{S}_n.$$

For a set $A \subseteq P$, write $\text{Share}_A(s)$ for the induced distribution on the shares held by A . For every authorized A there is a deterministic reconstruction function Rec_A .

The scheme has *perfect correctness* if every authorized set always reconstructs the secret. More generally, it has correctness error at most ε if

$$\Pr[\text{Rec}_A(\text{Share}_A(s)) \neq s] \leq \varepsilon,$$

for every $A \in \Gamma$ and every secret s .

For distributions X, Y on the same finite set, their statistical distance is

$$\Delta(X, Y) = \frac{1}{2} \sum_z |\Pr[X = z] - \Pr[Y = z]|.$$

A scheme has *pairwise statistical privacy parameter* δ if

$$\Delta(\text{Share}_B(s), \text{Share}_B(s')) \leq \delta,$$

for every unauthorized B and every pair of secrets s, s' . Perfect privacy is the case $\delta = 0$. This is the statistical-security convention used in [13].

When the secret space is $\mathbb{F}^{d_0}$ and participant i receives d_i field elements, put

$$d = \sum_{i=1}^n d_i, \quad \rho_i = \frac{d_i}{d_0}, \quad \rho_{\max} = \max_i \rho_i, \quad \rho_{\text{av}} = \frac{1}{n} \sum_i \rho_i = \frac{d}{nd_0}, \quad \rho_{\text{tot}} = \sum_i \rho_i = \frac{d}{d_0}.$$

We call ρ_i the *information ratio*, or the *normalized share size*, of participant i , and $\rho_{\max}$, ρ_{av} , ρ_{tot} the maximum, the average, and the total information ratio of the scheme.

2.2 Linear, multilinear, and degree-bounded reconstruction

A scheme over $\mathbb{F}$ is *multilinear* if the secret is a vector in $\mathbb{F}^{d_0}$ and there is a vector of independent uniform random field elements r such that the vector of all shares is a fixed $\mathbb{F}$ -linear map of (s, r) . Reconstruction is then linear. The case $d_0 = 1$ is usually called a *linear* secret-sharing scheme. We keep the distinction because the parameter d_0 is precisely the amortization dimension studied here.

A general scheme with shares in powers of $\mathbb{F}$ has *linear reconstruction* if, for every authorized set A and every secret coordinate, the reconstruction function is an $\mathbb{F}$ -linear form in the share coordinates held by A . It has *affine-linear reconstruction* if a constant term is also allowed. The sharing map itself need not be linear. More generally, the scheme has *degree- r reconstruction* if each secret coordinate is reconstructed by a polynomial of total degree at most r in the available share coordinates.

2.3 Multi-target monotone span programs

We use the strong-rejection form of multi-target monotone span programs from [7].

Definition 2.2 (MTMSP). A d_0 -target monotone span program over $\mathbb{F}$ consists of a matrix

$$M \in \mathbb{F}^{d \times b},$$

a labeling $\pi : [d] \rightarrow [n]$ of its rows by participants, and d_0 linearly independent target vectors spanning a subspace $T \leq \mathbb{F}^b$. We assume $1 \leq d_0 < b$; by convention 2.3 below this is no loss. For $A \subseteq P$, let M_A be the rows labeled by participants in A . The program

- *accepts* A if $T \subseteq \text{rowspan}(M_A)$;
- *rejects* A if $\text{rowspan}(M_A) \cap T = \{0\}$.

It computes Γ if it accepts exactly the authorized sets and rejects exactly the unauthorized sets.

Convention 2.3 (Zero columns). The strict inequality $d_0 < b$ follows [7]. Whenever it is not already satisfied, we append one zero column to the program matrix and to every target vector. This changes neither the row counts nor acceptance or rejection. We use this convention without further comment.

After an invertible change of coordinates in $\mathbb{F}^b$, we may and shall take

$$E = (I_{d_0} \ 0) \in \mathbb{F}^{d_0 \times b}, \quad T = \text{rowspan}(E).$$

If participant i labels d_i rows, then $d = \sum_i d_i$. The normalized total size is d/d_0 .

We record explicitly the direction of the span-program correspondence that will be used for lower bounds. It is here that perfect privacy yields the strong rejection condition.

Lemma 2.4 (From multilinear schemes to MTMSPs). *Let Π be a perfect multilinear secret-sharing scheme over $\mathbb{F}$ realizing Γ . Suppose the secret lies in $\mathbb{F}^{d_0}$ and participant i receives d_i field elements. Then there is a d_0 -target MTMSP computing Γ in which participant i labels exactly d_i rows.*

Proof. Write the full share vector as

$$Mx, \quad x = (s, r) \in \mathbb{F}^b,$$

where the first d_0 coordinates of x are the secret coordinates, and let

$$E = (I_{d_0} \ 0), \quad T = \text{rowspan}(E).$$

Apply convention 2.3 if necessary. The rows of M are already partitioned by participants, with exactly d_i rows belonging to participant i .

Let A be authorized. Perfect reconstruction implies that two vectors x, x' with $M_A x = M_A x'$ must have the same secret, and hence

$$\ker M_A \subseteq \ker E.$$

Taking annihilators gives $T = \text{rowspan}(E) \subseteq \text{rowspan}(M_A)$, so A is accepted.

Now let B be unauthorized and write

$$M_B = (S_B \ R_B),$$

according to secret and random coordinates. Since r is uniform, the view of B on secret s is uniform on the affine coset

$$S_B s + \text{colspan}(R_B).$$

Perfect privacy for all secrets is therefore equivalent to

$$\text{colspan}(S_B) \subseteq \text{colspan}(R_B).$$

Suppose $y \in \text{rowspan}(M_B) \cap T$. Then for some row vector a ,

$$y = aM_B = (u \ 0).$$

Thus $aR_B = 0$. Since every column of S_B lies in the column span of R_B , also $aS_B = 0$, and therefore $u = 0$. Hence $y = 0$ and

$$\text{rowspan}(M_B) \cap T = \{0\}.$$

Thus every unauthorized set is strongly rejected, proving the claim. $\square$

The converse construction, from an MTMSP to a perfect multilinear scheme, is the standard BBPT construction [7]; see also [9, Claim 4.15] for the participant-wise statement. In particular, the parameters ρ_i , ρ_{av} , and d/d_0 of section 2 may be read off from the row counts of the program.

For a monotone function f , let $s_{d_0}^{\mathbb{F}}(f)$ be the minimum number of rows of a d_0 -target MTMSP computing f . Such programs exist for every d_0 : take d_0 independent copies of a one-target program. Define the amortized span-program complexity

$$\text{amSP}_{\mathbb{F}}(f) = \inf_{d_0 \geq 1} \frac{s_{d_0}^{\mathbb{F}}(f)}{d_0}.$$

For $d_0 = 1$ this reduces to the ordinary monotone span-program model; we denote its minimum size by $\text{mSP}_{\mathbb{F}}(f)$.

2.4 The Razborov–Gál rank measure

Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$ be monotone, and write

$$U = f^{-1}(1), \quad V = f^{-1}(0).$$

For $i \in [n]$ define

$$R_i = \{(u, v) \in U \times V : u_i = 1, v_i = 0\}.$$

Monotonicity implies that the rectangles R_i cover $U \times V$: if $u \in U$ and $v \in V$, then $u \not\leq v$, so $u_i = 1, v_i = 0$ for some i .

If $A \in \mathbb{F}^{U \times V}$, let $A|_{R_i}$ be the matrix obtained by retaining $A[u, v]$ on R_i and putting zero elsewhere.

Definition 2.5 (Razborov–Gál rank measure). For $A \neq 0$, define

$$\mu_{\mathbb{F}}(f, A) = \frac{\text{rank}_{\mathbb{F}} A}{\max_{i \in [n]} \text{rank}_{\mathbb{F}}(A|_{R_i})}, \quad \mu_{\mathbb{F}}(f) = \max_{A \neq 0} \mu_{\mathbb{F}}(f, A).$$

For an access structure Γ , write $\mu_{\mathbb{F}}(\Gamma) = \mu_{\mathbb{F}}(f_{\Gamma})$.

The denominator in definition 2.5 is nonzero whenever $A \neq 0$, because the R_i cover every entry. This measure is a standard lower bound for ordinary monotone span programs; see [25, 17, 18, 24].

3 Rank lower bounds for multi-target span programs

We now prove the main theorem in full.

Theorem 3.1 (Rank measure lower-bounds normalized MTMSP size). *Let f be monotone and let a d_0 -target MTMSP over $\mathbb{F}$ compute f . If participant i labels d_i rows and $d = \sum_i d_i$, then for every nonzero $A \in \mathbb{F}^{U \times V}$,*

$$d_0 \text{rank } A \leq \sum_{i=1}^n d_i \text{rank}(A|_{R_i}).$$

Consequently,

$$\frac{d}{d_0} \geq \frac{\text{rank } A}{\max_i \text{rank}(A|_{R_i})}.$$

Maximizing over A and then over the target dimension gives

$$\boxed{\text{amSP}_{\mathbb{F}}(f) \geq \mu_{\mathbb{F}}(f).}$$

The proof uses two elementary linear-algebraic certificates associated with accepted and rejected inputs.

Lemma 3.2 (Simultaneous reconstruction). *For every $u \in U$ there is $L_u \in \mathbb{F}^{d_0 \times d}$ such that*

$$L_u M = E,$$

and every column of L_u corresponding to a row unavailable on u is zero.

Proof. Since u is accepted, every row of E belongs to $\text{rowspan}(M_u)$. Choose a reconstruction coefficient vector for each of the d_0 rows of E , pad it by zero coefficients outside the rows available on u , and stack the d_0 vectors. $\square$

Lemma 3.3 (Simultaneous rejection). *For every $v \in V$ there is $W_v \in \mathbb{F}^{b \times d_0}$ satisfying*

$$M_v W_v = 0, \quad E W_v = I_{d_0}.$$

Proof. Put $S = \text{rowspan}(M_v)$ and let

$$S^\perp = \{w \in \mathbb{F}^b : M_v w = 0\}.$$

Consider the linear map $E : S^\perp \rightarrow \mathbb{F}^{d_0}$. We claim that it is onto. If $y \in \mathbb{F}^{d_0}$ annihilates $E(S^\perp)$, then

$$E^T y \in (S^\perp)^\perp = S.$$

On the other hand, $E^T y \in T$. Since v is rejected, $S \cap T = \{0\}$, and hence $E^T y = 0$. As E has full row rank, $y = 0$. Thus the image $E(S^\perp)$ has trivial annihilator and equals $\mathbb{F}^{d_0}$.

For every standard basis vector $e_s \in \mathbb{F}^{d_0}$, choose $w_s \in S^\perp$ with $E w_s = e_s$. Then $W_v = [w_1 \cdots w_{d_0}]$ has the required properties. $\square$

Let $M_i \in \mathbb{F}^{d_i \times b}$ denote the row block labeled by participant i . Let $L_{u,i} \in \mathbb{F}^{d_0 \times d_i}$ be the corresponding block of columns in L_u . For $u \in U$, $v \in V$, define

$$B_i[u, v] = L_{u,i} M_i W_v \in \mathbb{F}^{d_0 \times d_0}.$$

The resulting block matrix B_i is the *tile of participant i* . When rank is taken, we flatten the tile by indexing its rows by $(u, a) \in U \times [d_0]$ and its columns by $(v, b) \in V \times [d_0]$.

Lemma 3.4 (Participant tiles). *For every participant i , its tile B_i satisfies:*

- (i) $B_i[u, v] = 0$ whenever $(u, v) \notin R_i$;
- (ii) $\sum_{i=1}^n B_i[u, v] = I_{d_0}$ for every $(u, v) \in U \times V$;
- (iii) $\text{rank } B_i \leq d_i$.

Proof. If $u_i = 0$, then the i -block of the reconstruction matrix vanishes: $L_{u,i} = 0$. If $v_i = 1$, every row of M_i belongs to M_v , so $M_i W_v = 0$ by lemma 3.3. This proves (i).

For (ii), use the block decomposition of $L_u M$:

$$\sum_i B_i[u, v] = \sum_i L_{u,i} M_i W_v = L_u M W_v = E W_v = I_{d_0}.$$

For (iii), vertically stack the matrices $L_{u,i}$ over $u \in U$ and horizontally concatenate the matrices $M_i W_v$ over $v \in V$:

$$P_i = \begin{pmatrix} L_{u_1,i} \\ L_{u_2,i} \\ \vdots \end{pmatrix} \in \mathbb{F}^{d_0|U| \times d_i}, \quad Q_i = \begin{pmatrix} M_i W_{v_1} & M_i W_{v_2} & \cdots \end{pmatrix} \in \mathbb{F}^{d_i \times d_0|V|}.$$

The flattened tile is exactly $B_i = P_i Q_i$. Therefore $\text{rank } B_i \leq d_i$. This also covers $d_i = 0$: the factorization is then through the zero-dimensional space and $B_i = 0$. $\square$

We use the standard inequality

$$\text{rank}(X \circ Y) \leq \text{rank}(X) \text{rank}(Y),$$

for the Hadamard product $X \circ Y$. It holds over every field: write each matrix as a sum of rank-one matrices and distribute the entrywise product.

Proof of theorem 3.1. Let $A_i = A|_{R_i}$ and let J_{d_0} be the all-ones $d_0 \times d_0$ matrix. Define a flattened matrix C_i blockwise by

$$C_i[u, v] = A[u, v] B_i[u, v].$$

Thus C_i is the tile of participant i weighted by the rank witness A . By lemma 3.4(i),

$$C_i = B_i \circ (A_i \otimes J_{d_0}).$$

Since the nonzero all-ones matrix J_{d_0} has rank one over every field, including when the characteristic divides d_0 ,

$$\text{rank}(A_i \otimes J_{d_0}) = \text{rank } A_i.$$

Therefore

$$\text{rank } C_i \leq \text{rank } B_i \text{rank}(A_i \otimes J_{d_0}) \leq d_i \text{rank } A_i.$$

On the other hand, lemma 3.4(ii) gives

$$\sum_i C_i = A \otimes I_{d_0}.$$

Hence

$$d_0 \text{rank } A = \text{rank}(A \otimes I_{d_0}) \leq \sum_i \text{rank } C_i \leq \sum_i d_i \text{rank } A_i \leq d \max_i \text{rank } A_i.$$

The first three terms prove the weighted inequality. The last inequality and division by $d_0 \max_i \text{rank } A_i$ give the normalized-size bound. Maximizing over $A \neq 0$, then taking the infimum over the target dimension, proves $\text{amSP}_{\mathbb{F}}(f) \geq \mu_{\mathbb{F}}(f)$. $\square$

Remark 3.5 (Relation to the BBPT argument). *In the unique-intersection application of BBPT [7, Sec. 4], the relevant participant-wise restrictions have rank one. Their argument therefore produces $D \otimes I_{d_0}$ and directly obtains a bound in terms of $\text{rank } D$. For a general Razborov–Gál witness A , the local matrices $A|_{R_i}$ need not have rank one. The tile factorization in lemma 3.4 and the Hadamard-product estimate retain these local ranks and give the factor $\text{rank}(A|_{R_i})$. This is the additional step needed to pass from the Gál–Pudlák criterion to the full rank measure.*

Remark 3.6 (The weighted profile bound). *The weighted inequality retains more information than the ratio bound. Define*

$$\nu_{\mathbb{F}}(f) = \inf \left\{ \sum_{i=1}^n x_i : x_i \geq 0 \text{ and } \text{rank } A \leq \sum_{i=1}^n x_i \text{rank}(A|_{R_i}) \text{ for every } A \neq 0 \right\}.$$

The normalized profile $x_i = d_i/d_0$ of every MTMSP is feasible, and hence

$$\frac{d}{d_0} \geq \nu_{\mathbb{F}}(f) \geq \mu_{\mathbb{F}}(f).$$

The second inequality follows by bounding every restricted rank by their maximum. We use only the simpler rank measure in this paper, but the profile bound may be strictly stronger.

3.1 A partial-specification variant

The remainder of this section is not used in sections 4 to 6 and may be skipped; it is referred to again only in section 7. The argument of theorem 3.1 applies verbatim when only selected positive and negative inputs are constrained.

Proposition 3.7 (Partial strong-rejection version). *Let $\mathcal{A}, \mathcal{B} \subseteq 2^P$. Suppose a labeled matrix with d_0 targets accepts every $X \in \mathcal{A}$ and strongly rejects every $Y \in \mathcal{B}$, i.e.,*

$$T \subseteq \text{rowspan}(M_X), \quad \text{rowspan}(M_Y) \cap T = \{0\}.$$

For $i \in [n]$, put

$$R_i(\mathcal{A}, \mathcal{B}) = \{(X, Y) \in \mathcal{A} \times \mathcal{B} : i \in X \setminus Y\}.$$

Then for every nonzero $Z \in \mathbb{F}^{\mathcal{A} \times \mathcal{B}}$,

$$\frac{d}{d_0} \geq \frac{\text{rank } Z}{\max_i \text{rank}(Z|_{R_i(\mathcal{A}, \mathcal{B})})}.$$

Proof. Repeat the proof of theorem 3.1 with U replaced by $\mathcal{A}$ and V replaced by $\mathcal{B}$. The denominator is nonzero: if $X \subseteq Y$ for some $X \in \mathcal{A}$ and $Y \in \mathcal{B}$, then $T \subseteq \text{rowspan}(M_X) \subseteq \text{rowspan}(M_Y)$, contradicting strong rejection. Hence every pair (X, Y) is covered by some $R_i(\mathcal{A}, \mathcal{B})$. $\square$

Remark 3.8. *Proposition 3.7 covers partial or ramp specifications in which designated forbidden sets are required to be perfectly private in the linear-algebraic sense. It does not apply to a weak rejection condition that permits an unauthorized set to learn a nonzero component of the target space.*

4 Exponential lower bounds for multilinear secret sharing

This section contains no new mathematics beyond bookkeeping: it feeds the rank witnesses of Pitassi and Robere into theorem 3.1. We restate their input and their lifting theorem as black boxes, because we need the constants and the thresholds to be independent of the field — a uniformity that their formulation, which fixes one field at a time, does not make explicit. Section 4.3 isolates that bookkeeping.

We first record that passing to a total extension is free.

Lemma 4.1 (Passing from a partial function to a total extension). *Let $h : \{0, 1\}^N \rightarrow \{0, 1, *\}$ be a partial monotone function and let f be a total monotone extension of h . Then*

$$\mu_{\mathbb{F}}(f) \geq \mu_{\mathbb{F}}(h),$$

where the rank measure of h is defined using its specified positive and negative inputs.

Proof. Let $U_0 = h^{-1}(1)$ and $V_0 = h^{-1}(0)$, and let Z be a rank witness on $U_0 \times V_0$. Since f extends h , its positive and negative input sets contain U_0 and V_0 , respectively. Extend Z to the full positive–negative matrix of f by putting zero in every new row and column. This operation preserves rank Z . For every input coordinate i , it also preserves the rank after masking by the rectangle $u_i = 1, v_i = 0$. The ratio witnessed by Z is therefore unchanged. $\square$

4.1 Two black boxes

For an unsatisfiable CNF formula C and a field $\mathbb{F}$, write $\text{NS}_{\mathbb{F}}(C)$ for the minimum degree of a Nullstellensatz refutation of C over $\mathbb{F}$, and $\text{PC}_{\mathbb{F}}(C)$ for the minimum degree of a polynomial-calculus refutation.

Proposition 4.2 (Proof-complexity input; Buss, Grigoriev, Impagliazzo, and Pitassi [15]). *For every prime p there are unsatisfiable CNF formulas $\text{MOD}_{p,t}$ of constant width, with $O(t)$ clauses in $O(t)$ variables, and a constant $\varepsilon_p > 0$ depending only on p , such that*

$$\text{NS}_{\mathbb{F}}(\text{MOD}_{p,t}) \geq \text{PC}_{\mathbb{F}}(\text{MOD}_{p,t}) \geq \varepsilon_p t,$$

for every field $\mathbb{F}$ with $\text{char } \mathbb{F} \neq p$. The constant ε_p does not depend on $\mathbb{F}$.

Proposition 4.3 (Lifting; Pitassi and Robere [24, Thms. 3.4–3.5]). *Let C be an unsatisfiable CNF of constant width with $O(t)$ clauses, let $\mathbb{F}$ be a field, and suppose $\text{NS}_{\mathbb{F}}(C) \geq \varepsilon t$ for some $\varepsilon > 0$. Let g be a good gadget of rank exceeding $2^{1/\varepsilon}$; a gadget of size depending only on ε suffices. Then the lifted partial monotone function $f_{C,g}$ on $N = O(t)$ Boolean inputs satisfies*

$$\mu_{\mathbb{F}}(f_{C,g}) \geq 2^{\Omega(t)},$$

where the implicit constants depend only on ε , the width of C , and the gadget g .

Proposition 4.3 is a lower bound on the rank measure itself, not merely on monotone span-program size; this is what makes it usable here.

4.2 The characteristic-separated families

Theorem 4.4 (Characteristic-separated rank witnesses; Pitassi–Robere, restated). *There are explicit families of total monotone functions $(f_{2,N})$ and $(f_{3,N})$, constants $\alpha_2, \alpha_3 > 0$, and thresholds N_2, N_3 with the following properties:*

(i) *for every finite field $\mathbb{F}$ of odd characteristic and every $N \geq N_2$,*

$$\mu_{\mathbb{F}}(f_{2,N}) \geq 2^{\alpha_2 N};$$

(ii) *for every finite field $\mathbb{F}$ of characteristic two and every $N \geq N_3$,*

$$\mu_{\mathbb{F}}(f_{3,N}) \geq 2^{\alpha_3 N}.$$

The constants and thresholds are independent of the particular finite field.

Proof. Fix $p \in \{2, 3\}$ and let $\varepsilon_p > 0$ be the constant of proposition 4.2. Choose a good gadget g_p of rank exceeding $2^{1/\varepsilon_p}$; by proposition 4.3 such a gadget of constant size exists, and it depends only on ε_p , hence only on p .

Let $\mathbb{F}$ be any field with $\text{char } \mathbb{F} \neq p$. By proposition 4.2, $\text{NS}_{\mathbb{F}}(\text{MOD}_{p,t}) \geq \varepsilon_p t$, so proposition 4.3 applied to $C = \text{MOD}_{p,t}$ and $g = g_p$ produces a partial monotone function $h_{p,t}$ on $N = \Theta(t)$ Boolean inputs with $\mu_{\mathbb{F}}(h_{p,t}) \geq 2^{c_p t}$, where $c_p > 0$ depends only on p . Let $f_{p,N}$ be the total monotone extension of $h_{p,t}$ used in [24, Theorem 7.3]; by lemma 4.1 the same witness gives $\mu_{\mathbb{F}}(f_{p,N}) \geq 2^{c_p t}$. Since $N = \Theta(t)$ with constants depending only on the fixed formulas and gadgets, there are $\alpha_p > 0$ and N_p with $\mu_{\mathbb{F}}(f_{p,N}) \geq 2^{\alpha_p N}$ for all $N \geq N_p$.

For (i) take $p = 2$: every field of odd characteristic has characteristic different from 2. For (ii) take $p = 3$: characteristic two is different from 3. All constants produced above depend only on p , whence the last sentence of the statement; see section 4.3. $\square$

Corollary 4.5. *Under the respective hypotheses of theorem 4.4,*

$$\text{amSP}_{\mathbb{F}}(f_{p,N}) \geq 2^{\alpha_p N} \quad (p \in \{2, 3\}).$$

Proof. Apply theorems 3.1 and 4.4. $\square$

4.3 Uniformity in the field

Pitassi and Robere argue over one field at a time, which is all their statements require. We need the stronger form $\exists \alpha \exists n_0 \forall \mathbb{F} \forall n \geq n_0$, because corollary 4.9 takes an infimum over all finite fields at each fixed n . Four points make this legitimate; none of them is a new theorem.

Characteristic, not field. The degree bounds of [15] depend on the field only through its characteristic. For $p = 2$ the formulas $\text{MOD}_{2,t}$ are the Tseitin systems of a fixed bounded-degree expander family, and the degree lower bound is stated in terms of the expansion parameter alone; it holds over every field of characteristic different from two. It genuinely fails in characteristic two, where polynomial calculus simulates Gaussian elimination on a linear system over $\text{GF}(2)$ and these systems have constant-degree refutations. This is why two families are needed, and it is the content of the linear separation between characteristics in [15]. For $p = 3$ we use the mod-3 counting principle, which is hard in every characteristic other than 3, in particular in characteristic two.

Polynomial calculus versus Nullstellensatz. The bounds of [15] are for polynomial-calculus degree, whereas proposition 4.3 consumes Nullstellensatz degree. The implication runs in the required direction: polynomial calculus simulates Nullstellensatz line by line, so $\text{PC}_{\mathbb{F}}(C) \leq \text{NS}_{\mathbb{F}}(C)$ for every C and $\mathbb{F}$, and a lower bound on $\text{PC}_{\mathbb{F}}$ yields the same lower bound on $\text{NS}_{\mathbb{F}}$. This is the chain displayed in proposition 4.2.

Invariance under field extension. At any fixed degree bound k , both Nullstellensatz and polynomial-calculus derivability are $\mathbb{F}$ -linear feasibility questions: one asks whether a fixed linear system, whose coefficients are the coefficients of the axioms and hence lie in the prime field, has a solution over $\mathbb{F}$. A linear system defined over $\mathbb{F}_p$ is solvable over $\mathbb{F}_{p^k}$ if and only if it is solvable over $\mathbb{F}_p$. Both degrees are therefore unchanged under finite extension, and a single ε_p serves an entire characteristic class.

Order of choice. The gadget in proposition 4.3 must have rank exceeding $2^{1/\varepsilon}$, so it can be chosen only after ε_p is known. By the three preceding points, ε_p depends on p alone; hence so do the gadget g_p and the resulting constants c_p , α_p , and N_p . This is the only place where the quantifier order matters, and it is discharged here once and for all.

4.4 One family over every finite field

To obtain one family that works over all finite fields, let $f_{2,N}$ and $f_{3,N}$ use disjoint sets of N variables and define

$$F_{2N}(x, y) = f_{2,N}(x) \vee f_{3,N}(y).$$

Each $f_{p,N}$ is nonconstant and monotone, and hence $f_{p,N}(0^N) = 0$. A zero restriction of an MTMSP is obtained by deleting the rows carrying variables fixed to zero. The target subspace is unchanged, so normalized MTMSP size cannot increase under such a restriction.

Theorem 4.6 (One family over every finite field). *There are an explicit monotone family (F_n) , a constant $\alpha > 0$, and a threshold n_0 such that, for every finite field $\mathbb{F}$ and every $n \geq n_0$,*

$$\text{amSP}_{\mathbb{F}}(F_n) \geq 2^{\alpha n}.$$

Proof. First let $n = 2N$ and use $F_{2N} = f_{2,N} \vee f_{3,N}$. If $\text{char } \mathbb{F} = 2$, fix the first block to zero and obtain $f_{3,N}$, to which theorem 4.4(ii) applies. If $\text{char } \mathbb{F} \neq 2$, fix the second block to zero and obtain $f_{2,N}$, to which theorem 4.4(i) applies. Thus

$$\text{amSP}_{\mathbb{F}}(F_{2N}) \geq 2^{\min(\alpha_2, \alpha_3)N}.$$

For odd $n = 2N + 1$, let F_n ignore its last input. Restricting that input to zero recovers F_{2N} , so the same lower bound holds. Taking

$$\alpha = \frac{1}{4} \min\{\alpha_2, \alpha_3\} \quad \text{and} \quad n_0 = 2 \max\{N_2, N_3\} + 1,$$

gives constants valid simultaneously for every finite field. $\square$

Theorem 4.7 (Exponential multilinear information ratio). *Let Γ_n be the access structure represented by F_n in theorem 4.6. With the constants $\alpha > 0$ and n_0 from that theorem, for every finite field $\mathbb{F}$ and every $n \geq n_0$, every perfect multilinear scheme realizing Γ_n satisfies*

$$\rho_{\text{av}} \geq \frac{2^{\alpha n}}{n} \quad \text{and} \quad \rho_{\text{max}} \geq \frac{2^{\alpha n}}{n}.$$

Proof. Let the corresponding MTMSP have d_0 targets and d_i rows labeled by participant i ; it exists by lemma 2.4. By theorem 4.6,

$$\frac{d}{d_0} = \sum_{i=1}^n \frac{d_i}{d_0} \geq 2^{\alpha n}.$$

Therefore

$$\rho_{\text{av}} = \frac{d}{nd_0} \geq \frac{2^{\alpha n}}{n} = 2^{\Omega(n)}.$$

Since $\rho_{\text{max}} \geq \rho_{\text{av}}$, the same lower bound holds for the maximum ratio. $\square$

Remark 4.8 (Attribution for the scalar case). *An all-finite-fields exponential lower bound for one-element linear schemes, with one constant valid for every finite field, is already recorded in [13, extended version, Theorem 4.10], with attribution there to Pitassi–Robere and to Robere’s thesis [27]. Theorem 4.6 claims no novelty over that statement for $d_0 = 1$: it records the corresponding uniformity at the level of the rank witnesses rather than of share size, and makes explicit the quantifier order that target amortization requires. What is new is theorem 3.1, which shows that the same obstruction persists for every target dimension.*

Corollary 4.9 (Worst-case asymptotic order). *Let*

$$R_{\text{multi}}(n) = \max_{\Gamma \text{ on } n \text{ participants}} \inf_{\substack{\mathbb{F} \text{ finite, } d_0 \geq 1, \\ \Pi \text{ a perfect } d_0\text{-multilinear} \\ \text{scheme for } \Gamma \text{ over } \mathbb{F}}} \rho_{\max}(\Pi).$$

Then, with $\alpha > 0$ and n_0 as in theorem 4.6, for every $n \geq n_0$,

$$\boxed{\frac{2^{\alpha n}}{n} \leq R_{\text{multi}}(n) \leq 2^{\lceil n/2 \rceil - 1} + 1, \quad \text{so} \quad R_{\text{multi}}(n) = 2^{\Theta(n)}.$$

Proof. The lower bound is theorem 4.7. For the upper bound, Nir realizes every n -party access structure by a linear scheme for a one-bit secret whose maximum share size is $2^{\lceil n/2 \rceil - 1} + 1$ bits [22]; over $\text{GF}(2)$ this is a multilinear scheme with $d_0 = 1$, so $\rho_{\max} = \max_i d_i$ is bounded by the same quantity. (The classical monotone-formula construction already gives $2^{O(n)}$; see, e.g., [8, 9].) $\square$

5 Arbitrary sharing with affine-linear reconstruction

We next show that the rank-measure obstruction does not require a linear sharing algorithm. This section is based on the dual construction of Beimel, Othman, and Peter [13, extended version, Sec. 10]; we include the construction and proof because the exact hypotheses are important for long secrets and statistical privacy.

Definition 5.1 (Admissible scheme). A scheme Π over a finite field $\mathbb{F}$ is *admissible* if its secret domain $\mathcal{S}_0 \subseteq \mathbb{F}^{d_0}$ satisfies

$$\text{span}(\mathcal{S}_0 - \mathcal{S}_0) = \mathbb{F}^{d_0},$$

the scheme has perfect correctness and affine-linear reconstruction, and its pairwise statistical privacy parameter satisfies $\delta < 1$. No restriction is placed on the sharing algorithm.

Construction 5.2 (The dual program). Let Π be a scheme over $\mathbb{F}$ with secret domain $\mathcal{S}_0 \subseteq \mathbb{F}^{d_0}$. Participant i receives d_i field elements and $d = \sum_i d_i$. Index the d share coordinates by a set C , partitioned as $C = C_1 \sqcup \dots \sqcup C_n$, and denote a full share vector by $\sigma(s, r) \in \mathbb{F}^d$.

Suppose reconstruction is affine-linear. For every minimal authorized set $A \in \min \Gamma$ and every secret coordinate $j \in [d_0]$, choose a vector $\lambda^{A,j} \in \mathbb{F}^d$ supported on the share coordinates held by A , together with a constant $b_{A,j} \in \mathbb{F}$, such that

$$\langle \lambda^{A,j}, \sigma(s, r) \rangle + b_{A,j} = s_j,$$

for all secrets s and all dealer randomness r .

Form a matrix

$$\Lambda \in \mathbb{F}^{d \times (d_0 |\min \Gamma|)},$$

whose columns are the vectors $\lambda^{A,j}$, indexed by pairs (A, j) . For $j \in [d_0]$, define the target vector τ_j by

$$\tau_j[(A, j')] = \begin{cases} 1, & j' = j, \\ 0, & j' \neq j. \end{cases}$$

The τ_j are linearly independent because $\min \Gamma$ is nonempty. Label row $t \in C_i$ of Λ by participant i . If $|\min \Gamma| = 1$, apply convention 2.3.

Lemma 5.3 (Dual MTMSP from affine-linear reconstruction). *Assume perfect correctness,*

$$\text{span}(\mathcal{S}_0 - \mathcal{S}_0) = \mathbb{F}^{d_0},$$

and the following weak non-disjointness privacy condition:

for every unauthorized B and every $s, s' \in \mathcal{S}_0$, the supports of $\text{Share}_B(s)$ and $\text{Share}_B(s')$ are not disjoint.

Then the pair $(\Lambda, \{\tau_1, \dots, \tau_{d_0}\})$ of construction 5.2 is a d_0 -target MTMSP for Γ^ with exactly d rows.*

Proof. Let $T = \text{span}\{\tau_1, \dots, \tau_{d_0}\}$.

First consider $X \notin \Gamma^*$. Then $P \setminus X \in \Gamma$. Choose a minimal authorized subset $A \subseteq P \setminus X$. Suppose a linear combination of the rows of Λ labeled by X equals a target vector $\sum_j \alpha_j \tau_j$. Examine the coordinate (A, j') . Every reconstruction vector $\lambda^{A,j'}$ is supported inside A , so every row labeled outside A has zero in this coordinate. Hence $\alpha_{j'} = 0$ for every j' . Thus

$$\text{rowspan}(\Lambda_X) \cap T = \{0\},$$

which is strong rejection.

Now let $X \in \Gamma^*$, so $B = P \setminus X \notin \Gamma$. Suppose $T \not\subseteq \text{rowspan}(\Lambda_X)$. By the same annihilator argument used in lemma 3.3, there are a nonzero target $\tau = \sum_j \alpha_j \tau_j$ and a vector w such that

$$\Lambda_X w = 0, \quad \tau w = 1.$$

Write $w = (w_{A,j})_{A,j}$ and put

$$\mu = \sum_{A,j} w_{A,j} \lambda^{A,j} \in \mathbb{F}^d.$$

The equation $\Lambda_X w = 0$ says that μ vanishes on every share coordinate held by X , so μ is supported inside the unauthorized set B . On every valid sharing,

$$\langle \mu, \sigma(s, r) \rangle = \sum_{A,j} w_{A,j} (s_j - b_{A,j}) = \sum_j \beta_j s_j - \gamma =: \lambda(s),$$

where

$$\beta_j = \sum_A w_{A,j}, \quad \gamma = \sum_{A,j} w_{A,j} b_{A,j}.$$

Moreover,

$$1 = \tau w = \sum_j \alpha_j \sum_A w_{A,j} = \sum_j \alpha_j \beta_j,$$

so $\beta = (\beta_j)_j$ is nonzero. Since $\text{span}(\mathcal{S}_0 - \mathcal{S}_0) = \mathbb{F}^{d_0}$, choose $s, s' \in \mathcal{S}_0$ with $\beta \cdot (s - s') \neq 0$. The affine constant γ cancels, and hence $\lambda(s) \neq \lambda(s')$.

The fixed linear functional μ , supported on B , takes the constant value $\lambda(s)$ on every point in the support of $\text{Share}_B(s)$ and the different value $\lambda(s')$ on every point in the support of $\text{Share}_B(s')$. The two supports are therefore disjoint, contradicting the hypothesis. Hence $T \subseteq \text{rowspan}(\Lambda_X)$, and X is accepted. $\square$

Remark 5.4 (Sharpness of the hypotheses). *Pairwise statistical privacy with any $\delta < 1$ implies the non-disjointness condition, because two finite distributions have disjoint supports if and only if their statistical distance is one. The threshold $\delta < 1$ is therefore sharp for pairwise statistical distance. The spanning assumption is exact for this proof as well: it says precisely that every nonzero linear functional is nonconstant on the secret domain. In particular it holds for $\mathcal{S}_0 = \mathbb{F}^{d_0}$.*

We also record the elementary duality invariance of the rank measure.

Lemma 5.5 (Duality invariance). *For every access structure Γ as defined in section 2 and every field $\mathbb{F}$,*

$$\mu_{\mathbb{F}}(\Gamma^*) = \mu_{\mathbb{F}}(\Gamma).$$

Proof. Let U and V be the authorized and unauthorized sets of Γ . The authorized sets of Γ^* are $P \setminus V$, and its unauthorized sets are $P \setminus U$. Given $A \in \mathbb{F}^{U \times V}$, define

$$A^*[P \setminus Y, P \setminus X] = A[X, Y].$$

Up to row and column relabeling, $A^* = A^T$, so $\text{rank } A^* = \text{rank } A$. Moreover,

$$i \in (P \setminus Y) \setminus (P \setminus X) \iff i \in X \setminus Y.$$

Thus each coordinate mask is carried to the corresponding mask by transpose and permutations, and every masked rank is preserved. The map $A \mapsto A^*$ is a bijection, so the maxima defining the two rank measures are equal. $\square$

The following theorem combines the dual construction of Beimel, Othman, and Peter with the rank bound of theorem 3.1.

Theorem 5.6 (Rank lower bound for affine-linear reconstruction). *Let Π be an admissible scheme realizing Γ over $\mathbb{F}$. If the shares contain d field elements in total, then*

$$\boxed{\rho_{\text{tot}} = \frac{d}{d_0} \geq \mu_{\mathbb{F}}(\Gamma).}$$

Proof. By lemma 5.3, whose hypotheses hold by definition 5.1 and remark 5.4, the reconstruction vectors give a d_0 -target MTMSP for Γ^* with exactly d rows. By theorem 3.1,

$$\frac{d}{d_0} \geq \mu_{\mathbb{F}}(\Gamma^*).$$

Apply lemma 5.5. $\square$

Corollary 5.7 (Exponential lower bound with nonlinear sharing). *For the explicit family (Γ_n) of theorem 4.7, over every finite field every admissible scheme has*

$$\rho_{\text{tot}} = \frac{d}{d_0} \geq 2^{\Omega(n)}, \quad \rho_{\text{av}} \geq 2^{\Omega(n)}, \quad \rho_{\text{max}} \geq 2^{\Omega(n)}.$$

Proof. Combine theorems 4.6 and 5.6. Dividing the total normalized size by n changes only a polynomial factor, which is absorbed by the exponential bound. $\square$

Remark 5.8 (Comparison with BOP). *Beimel, Othman, and Peter's Construction 10.6 and Claim 10.8 in the extended version [13] give the same dual MTMSP for their displayed $(0, 1/2)$ statistical-security parameter. Their Definition 2.4 uses pairwise statistical distance, and their proof only needs to exclude distance one. Thus the argument applies to every $\delta < 1$. Their extended-version Theorem 10.9 also converts such linear-reconstruction schemes to linear sharing and reconstruction without changing d or d_0 under its stated hypotheses. The affine extension above is harmless because the reconstruction constants cancel when two secrets are compared.*

6 Degree-bounded reconstruction and small correctness error

6.1 Constant-degree reconstruction

Lemma 6.1 (Degree reduction; Beimel, Othman, and Peter [13, extended version, Lemma 4.7]). *Let Π be a scheme with a d_0 -dimensional secret, degree- r reconstruction, and d field elements in all shares together. Then Π can be transformed into a scheme with affine-linear reconstruction, the same secret dimension, and the same correctness and statistical-privacy parameters, in which each participant receives at most $K_r d^r$ field elements, where K_r depends only on r .*

Here d is the total original share dimension, whereas the new bound is participant-wise. In the BOP proof, the transformed view of an unauthorized set is obtained from its original view by a secret-independent randomized map; hence pairwise statistical distance does not increase. The affine formulation is necessary because a degree-one polynomial may have a constant term.

Theorem 6.2 (Long secrets with degree- r reconstruction). *There are constants $\alpha > 0$ and n_0 , and for every fixed $r \geq 2$ a constant $K_r > 0$, such that the explicit family (Γ_n) of theorem 4.7 has the following property. Let $n \geq n_0$ and let Π be a perfectly correct scheme over a finite field with secret space $\mathbb{F}^{d_0}$, pairwise statistical privacy parameter $\delta < 1$, degree- r reconstruction, and d total share field elements. Then*

$$\rho_{\text{tot}} = \frac{d}{d_0} \geq K_r^{-1/r} 2^{\alpha n/r} d_0^{-(1-1/r)}.$$

Consequently, for every fixed r , if $d_0 = 2^{o(n)}$, then

$$\frac{d}{d_0} \geq 2^{\Omega(n)}, \quad \rho_{\text{av}} \geq 2^{\Omega(n)}, \quad \rho_{\text{max}} \geq 2^{\Omega(n)}.$$

Proof. Apply lemma 6.1. The resulting scheme is admissible and has the same secret dimension d_0 . By corollary 5.7, after decreasing α if necessary, some participant in the transformed scheme receives at least

$$d_0 2^{\alpha n},$$

field elements, while the participant-wise upper bound of lemma 6.1 is $K_r d^r$. Therefore

$$K_r d^r \geq d_0 2^{\alpha n},$$

and hence

$$d \geq K_r^{-1/r} d_0^{1/r} 2^{\alpha n/r}.$$

Dividing by d_0 gives the displayed bound. If $d_0 = 2^{o(n)}$ and r is fixed, its logarithm is

$$\frac{\alpha}{r}n - o(n) - O_r(1) = \Omega(n).$$

The average ratio is $d/(nd_0)$ and the maximum is at least the average; division by n is absorbed by the exponential lower bound. $\square$

Remark 6.3 (Range of secret dimensions covered). *Quantitatively, if $d_0 = 2^{\gamma n}$, then the displayed bound of theorem 6.2 remains exponential exactly when $\gamma < \alpha/(r-1)$. For $r = 1$, theorem 5.6 is the sharper statement and has no dependence on d_0 at all.*

6.2 Conditioning away a small correctness error

For completeness, we record a simple reduction from sufficiently small reconstruction error to perfect correctness. The statement is useful because it preserves the reconstruction functions, and therefore preserves linearity or degree bounds.

Lemma 6.4 (Conditioning lemma). *Let Π be an (ε, δ) scheme for Γ with a finite secret domain and fixed deterministic reconstruction functions. Put*

$$a = |\Gamma|\varepsilon.$$

If $a < 1$, there is a perfectly correct scheme with the same share alphabets and the same reconstruction functions whose pairwise privacy parameter is at most

$$\delta + 2a.$$

In particular, $a \leq 2^n \varepsilon$.

Proof. Fix a secret s . Let E_s be the event, over the dealer randomness, that every authorized set reconstructs s correctly. By the union bound,

$$\Pr[E_s^c] \leq |\Gamma|\varepsilon = a.$$

Condition the full share distribution for secret s on E_s . The resulting sharing is perfectly correct. If D_s is the original full-share distribution and $D'_s = D_s \mid E_s$, then

$$\Delta(D_s, D'_s) = \Pr[E_s^c] \leq a.$$

Statistical distance cannot increase under projection. Hence, for every unauthorized B and secrets s, s' ,

$$\begin{aligned} \Delta(D'_{s,B}, D'_{s',B}) &\leq \Delta(D'_{s,B}, D_{s,B}) + \Delta(D_{s,B}, D_{s',B}) + \Delta(D_{s',B}, D'_{s',B}) \\ &\leq a + \delta + a. \end{aligned}$$

The reconstruction functions are unchanged.

To fit the standard model with one finite uniform random domain for all secrets, suppose the original randomness set is finite. Each conditioning event corresponds to a nonempty subset E_s of that set. Let L be a common multiple of the numbers $|E_s|$. A uniform variable on $[L]$ can be mapped, separately for each secret, onto E_s with equal-size fibers. This samples the conditioned distribution exactly and uses a common uniform randomness domain. $\square$

Remark 6.5 (Comparison with BOP’s conditioning argument). *BOP extended-version Claim 10.10 treats one-bit secrets and obtains privacy parameter $2\delta + 2^{n+1}\varepsilon$ under the stronger condition $\varepsilon < 2^{-n-2}$. Lemma 6.4 applies to every finite secret domain and gives $\delta + 2|\Gamma|\varepsilon \leq \delta + 2^{n+1}\varepsilon$. The improvement from 2δ to δ comes from conditioning the full sharing distribution directly and applying the triangle inequality. Under a distributional definition of sharing this construction is immediate; the last paragraph of the proof supplies a common uniform randomness space when that presentation is required.*

Corollary 6.6. *Let Π have affine-linear reconstruction and secret domain $\mathcal{S}_0 \subseteq \mathbb{F}^{d_0}$ with $\text{span}(\mathcal{S}_0 - \mathcal{S}_0) = \mathbb{F}^{d_0}$. Suppose its correctness error is ε and its pairwise statistical privacy parameter is δ . If*

$$\delta + 2|\Gamma|\varepsilon < 1,$$

then

$$\frac{d}{d_0} \geq \mu_{\mathbb{F}}(\Gamma).$$

The same preliminary conditioning can be applied before theorem 6.2 for degree- r reconstruction.

Proof. Apply lemma 6.4. The conditioned scheme has perfect correctness, privacy parameter below one, the same share lengths, and the same reconstruction maps. Then apply theorem 5.6. $\square$

7 Discussion and open problems

Theorem 3.1 identifies a class of one-dimensional lower bounds that is preserved when the target dimension is arbitrary. Not every lower-bound method has this property, and the distinction is worth stating explicitly. Counting arguments can show that most one-target programs are large, but a longer target changes the objects being counted and may destroy the normalized bound; indeed there are access structures for which one-dimensional linear schemes are expensive while long-secret schemes have small normalized complexity [5, 9], and those separations are against counting bounds. A rank argument instead supplies a specific algebraic obstruction, and for the Razborov–Gál measure we have

$$\mu_{\mathbb{F}}(f) \leq \text{amSP}_{\mathbb{F}}(f) \leq \text{mSP}_{\mathbb{F}}(f).$$

Thus every function with large rank measure remains hard for multi-target span programs after normalization by the target dimension. Read in the other direction, the same inequality turns the efficient long-secret schemes of Applebaum and Arkis [5] into an upper bound on the rank measure of the access structures they realize: those structures must have small $\mu_{\mathbb{F}}$ over every field. This is a consequence of theorem 3.1, not a tension with it.

The exponential consequence answers Beimel’s Question 10.4 and settles the worst-case multilinear information ratio at the $2^{\Theta(n)}$ scale, but several quantitative and structural questions remain.

The exponent. Write the worst-case multilinear exponent as $\limsup_n n^{-1} \log_2 R_{\text{multi}}(n)$. Our lower bound uses the Pitassi–Robere characteristic-separation families and yields only some constant $\alpha > 0$, whereas the upper bound of corollary 4.9 is $1/2$; the exponent therefore lies in $[\alpha, \frac{1}{2}]$ and is not determined here. The corresponding question for $d_0 = 1$ has just been settled: Nir’s linear scheme has maximum share size $2^{\lceil n/2 \rceil - 1} + 1$ for a one-bit secret, and a counting lower bound for monotone span programs shows that almost all access structures require linear share size $2^{n/2 - o(n)}$,

so the worst-case linear exponent is exactly $1/2$ [22]. It is worth being precise about what this does and does not say here. The matching bound on the linear side is a *counting* bound, and counting bounds are exactly the ones that need not survive amortization; nothing in [22] excludes a multilinear exponent strictly below $1/2$. Deciding whether amortization helps in the worst case — equivalently, whether the multilinear exponent is smaller than the linear one — is the sharp form of the question left open here, and answering it downwards would require constructions rather than the obstruction studied in this paper. That such separations do occur once linearity is relaxed is already visible one degree up: the same work gives quadratic schemes of total share size $2^{0.4995n+o(n)}$, and general schemes of total share size $2^{0.496n+o(n)}$, both below $2^{n/2}$, so the quadratic and linear exponents are already separated [22].

Arbitrarily long secrets with degree- r reconstruction. For $r > 1$, theorem 6.2 loses the factor $d_0^{1-1/r}$, and by remark 6.3 it answers BOP’s Question 1.1 for every subexponential secret length but not for arbitrary long secrets. Removing this loss would settle the remaining constant-degree case. One possible route is to find a degree- r analogue of the rank measure that is itself stable under target amortization, rather than first reducing degree r to degree one.

Beyond strong rejection. The proof of theorem 3.1 uses the condition $\text{rowspan}(M_B) \cap T = \{0\}$ for forbidden sets. Proposition 3.7 shows that intermediate coalitions may be left unspecified, but a forbidden coalition that learns a nonzero target component breaks the simultaneous rejection certificate. Understanding rank methods for weaker forms of leakage or ramp privacy is a separate problem. Partial and statistical security notions such as those studied in [19] provide a natural language for this question.

Rank measure versus other algebraic obstructions. The theorem transfers every lower bound that is already visible to $\mu_{\mathbb{F}}$, but it says nothing about scalar linear lower bounds obtained by counting or by measures not controlled by rank. It is therefore natural to ask for explicit access structures whose rank measure is small but whose multilinear complexity is large. Such examples would require lower-bound mechanisms genuinely beyond the rank-measure framework.

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